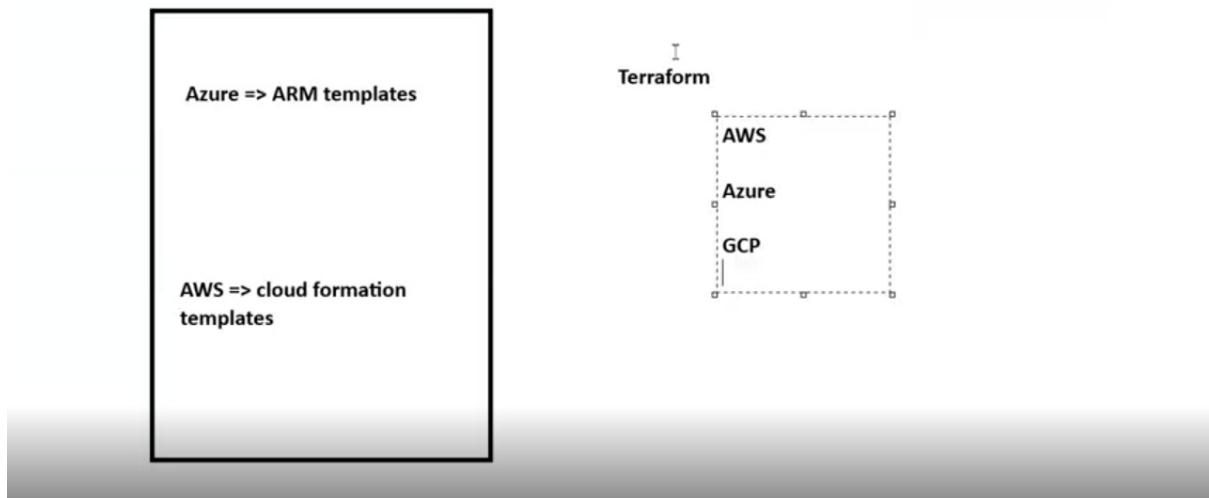


1.Terraform Azure



Terraform (iac)

Advantages:

- it supports multiple providers
- it is idempotent : if you run same script it wont affect present infra as it holds "state file"
- can dry run the script => you can plan the execution
- split configuration into multiple files
- manually created resources can be imported into terraform

terraform file extension

.tf

two major blocks

provider block : on which provider you are working

resource block : which resource "RG1" "vnet"

```
provider {
  azurearm
}

Resource {
  RG, Vnet, Subnet
}
```



Terraform init : it will download all provider related plugins [.terraform]

terraform validate : it will validate code syntax

terraform plan : it will show the plan execution

terraform apply : it will create the resources on cloud

terraform destroy: it will delete all the resource

terraform fmt: it will format your code / organize your code

- ✓ Download Terraform
- ✓ Create folder on **C-drive** and paste **terraform file**, then open terminal there.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Terraform> dir

Directory: C:\Terraform

Mode                LastWriteTime         Length Name
----                -
-a-----         03/03/2026    09:35      102918280 terraform.exe
```

```

PS C:\Terraform> .\terraform -- version
Usage: terraform [global options] <subcommand> [args]

The available commands for execution are listed below.
The primary workflow commands are given first, followed by
less common or more advanced commands.

Main commands:
  init          Prepare your working directory for other commands
  validate      Check whether the configuration is valid
  plan          Show changes required by the current configuration
  apply         Create or update infrastructure
  destroy       Destroy previously-created infrastructure

All other commands:
  console       Try Terraform expressions at an interactive command prompt
  fmt          Reformat your configuration in the standard style
  force-unlock  Release a stuck lock on the current workspace
  get          Install or upgrade remote Terraform modules
  graph        Generate a Graphviz graph of the steps in an operation
  import       Associate existing infrastructure with a Terraform resource
  login        Obtain and save credentials for a remote host
  logout       Remove locally-stored credentials for a remote host
  metadata     Metadata related commands
  modules      Show all declared modules in a working directory
  output       Show output values from your root module
  providers    Show the providers required for this configuration
  query        Search and list remote infrastructure with Terraform
  refresh      Update the state to match remote systems
  show         Show the current state or a saved plan
  stacks       Manage HCP Terraform stack operations
  state        Advanced state management
  taint        Mark a resource instance as not fully functional
  test         Execute integration tests for Terraform modules
  untaint      Remove the 'tainted' state from a resource instance
  version      Show the current Terraform version
  workspace    Workspace management

Global options (use these before the subcommand, if any):
  -chdir=DIR    Switch to a different working directory before executing the
                given subcommand.
  -help         Show this help output or the help for a specified subcommand.
  -version      An alias for the "version" subcommand.
PS C:\Terraform>

```

- ✓ Below it will show the version bcz we are opening at terraform folder below.

```

PS C:\terraform> .\terraform --version
Terraform v1.14.6
on windows_amd64
PS C:\terraform>

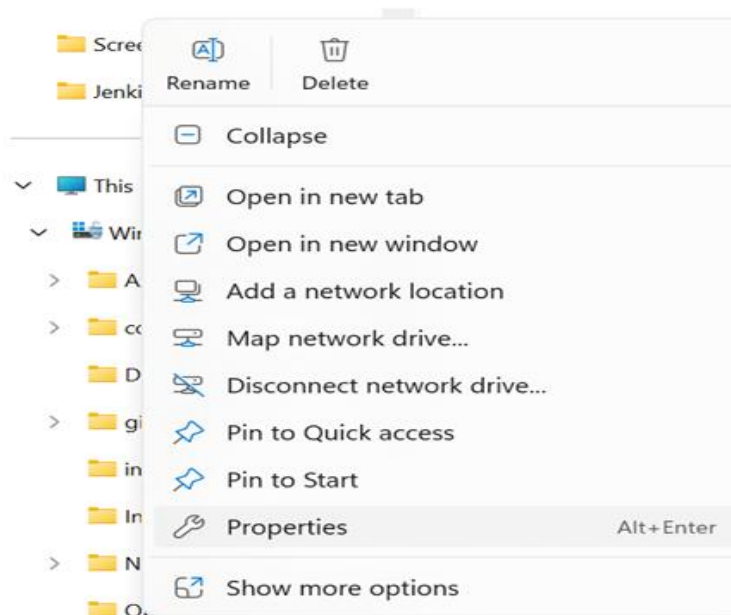
```

- ✓ Below it shows error bcz we changed directory to C:\
- ✓ Terraform.exe file is available in terraform folder.
- ✓ **terraform.exe** is not available in **C- drive**. That's why error shows below.

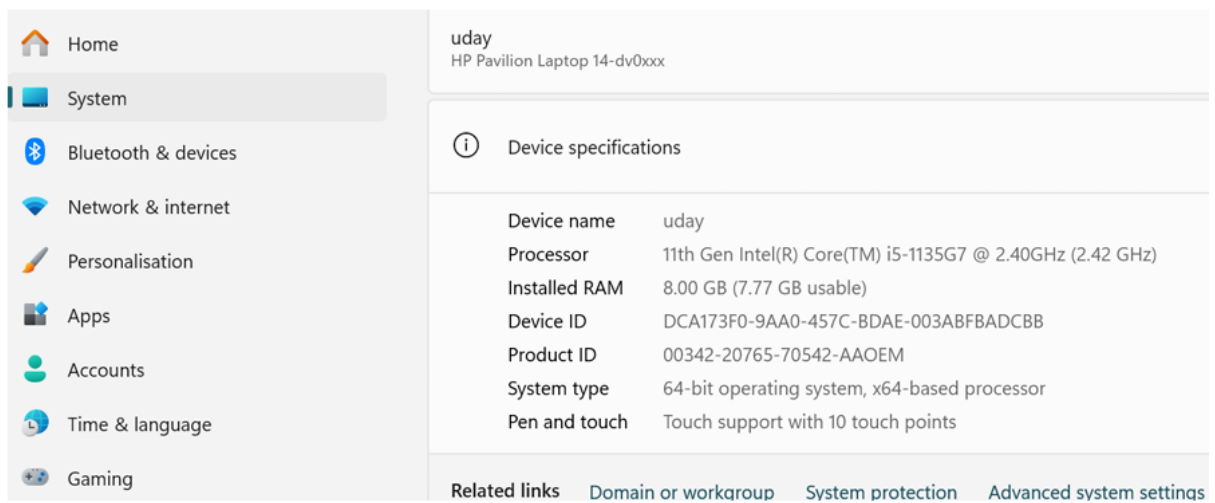
```
PS C:\terraform> .\terraform --version
Terraform v1.14.6
on windows_amd64
PS C:\terraform> cd ..
PS C:\> .\terraform --version
.\terraform : The term '.\terraform' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the
spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ .\terraform --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (.\terraform:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\>
```

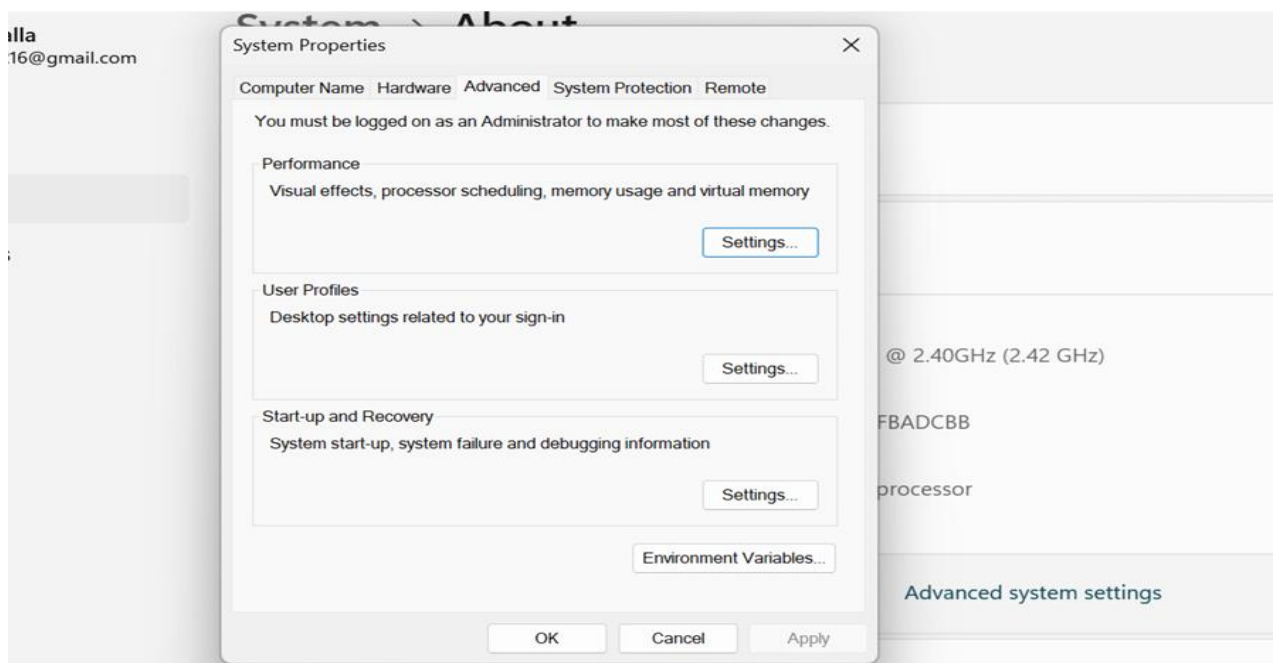
- ✓ To avoid this error, we should configure ENV
- ✓ Let's configure Environment Variable, to run .exe from anywhere in the system.
- ✓ For that click on This PC below



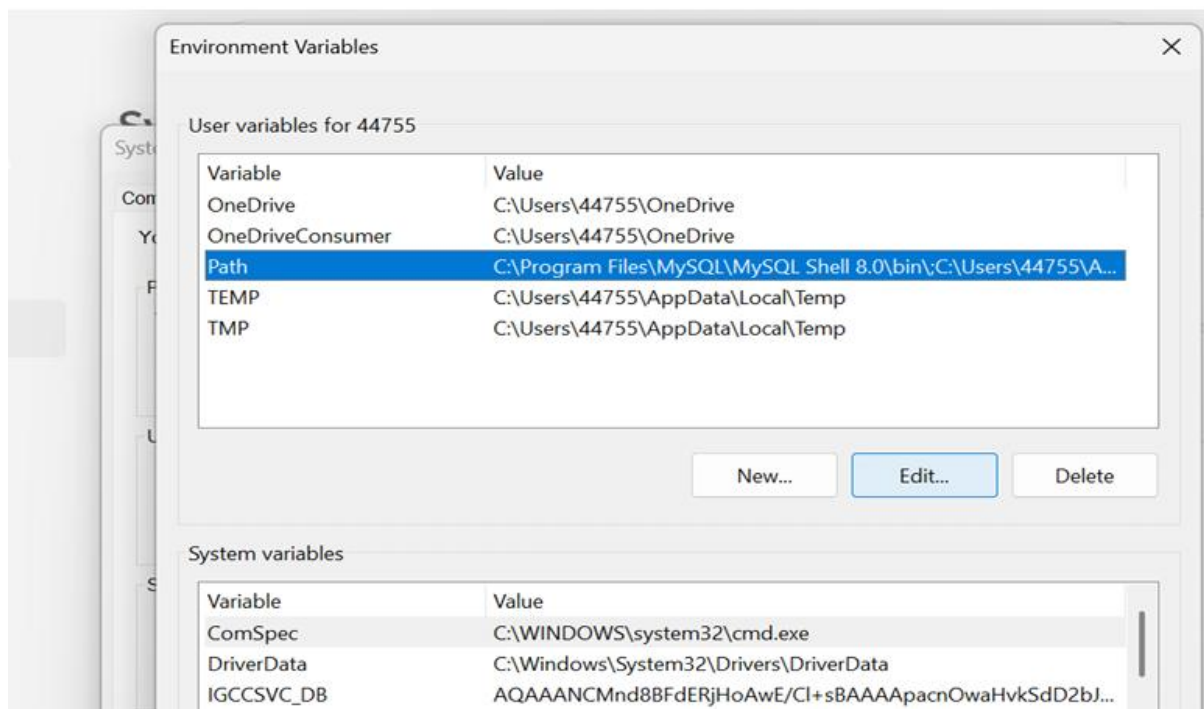
✓ System → Advanced System settings → Click



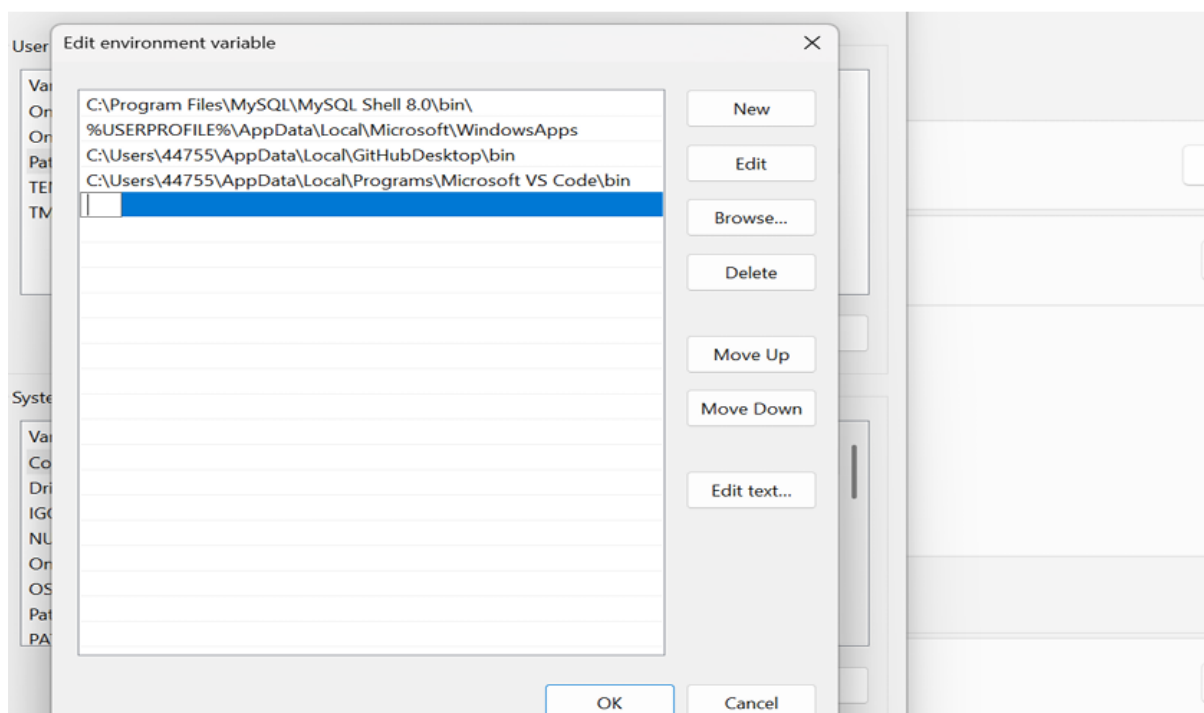
✓ Click on below **Environment Variables**



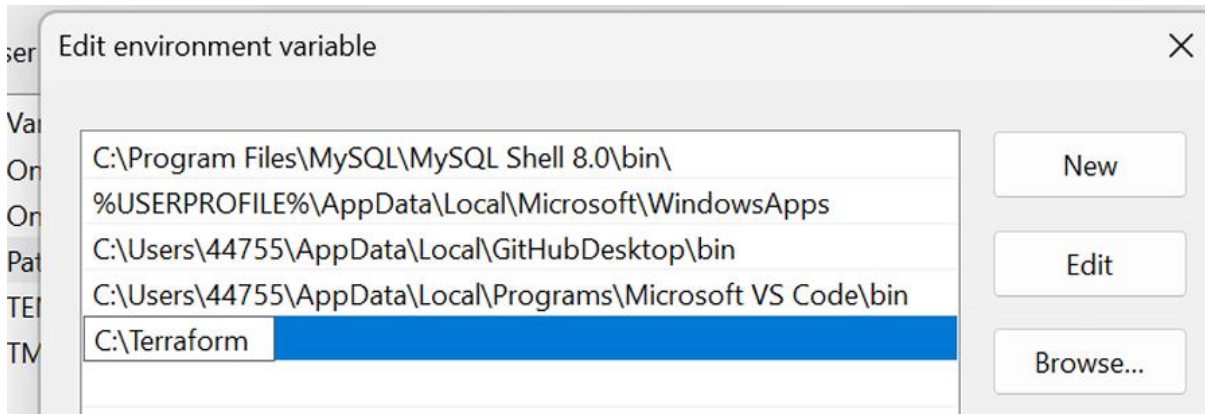
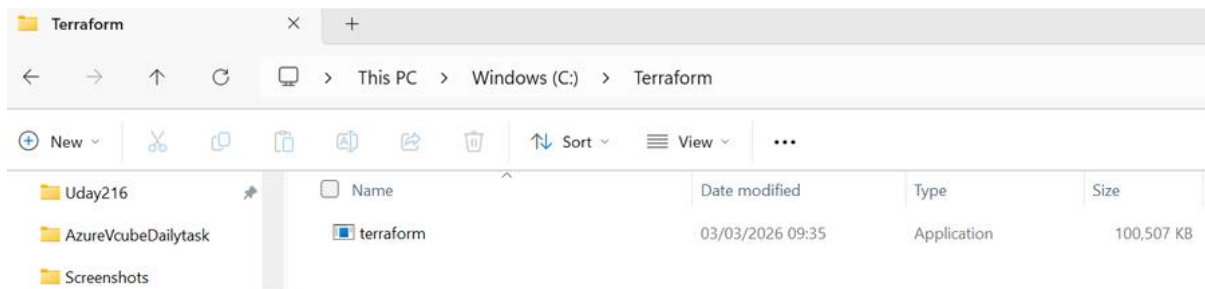
- ✓ Select path and edit below.



- ✓ Select new and paste the path below.



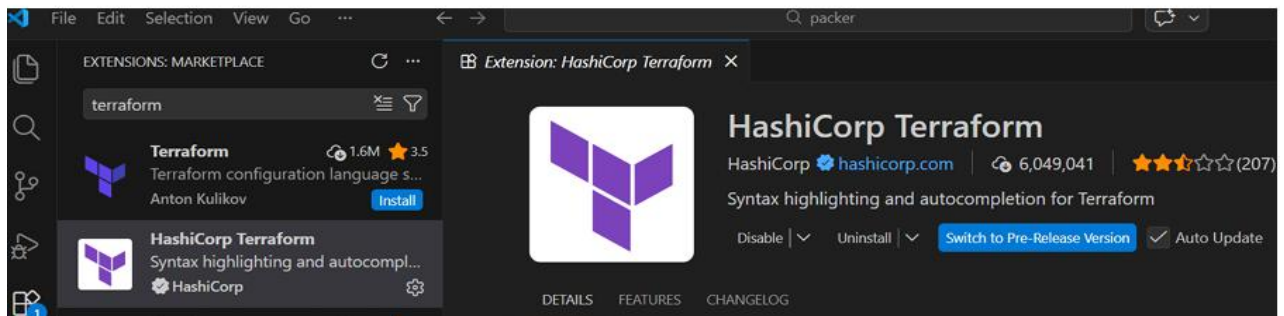
- ✓ Copy this below path in ENV



- ✓ Instead of pasting .exe in each folder, this .exe should run from any path so that we configured ENV.
- ✓ Below we executed from C-drive and we can run from anywhere because of environment variables.

```
PS C:\Terraform> terraform --version
Terraform v1.14.6
on windows_amd64
PS C:\Terraform> cd ..
PS C:\> terraform --version
Terraform v1.14.6
on windows_amd64
PS C:\> Terraform --version
Terraform v1.14.6
on windows_amd64
```

- Now let's configure terraform from VS Code for that Install VS code and within that install terraform extension (this will suggest us code when we are writing).



✓ Terraform major blocks

terraform file extension
.tf

two major blocks

provider block : on which provider you are working

resource block : which resource "RG1" "vnet"

✓ Below these are the major commands

1. terraform init => it will initialize terraform and it will download provider related plugins [.terraform]
2. terraform validate => it validates code syntax
3. terraform plan : it execute plan
4. terraform apply : it will create resources on cloud
5. Terraform destroy: it will destroy all resources
5. terraform fmt : it will organize your code
7. terraform taint: it will re-create same resource again

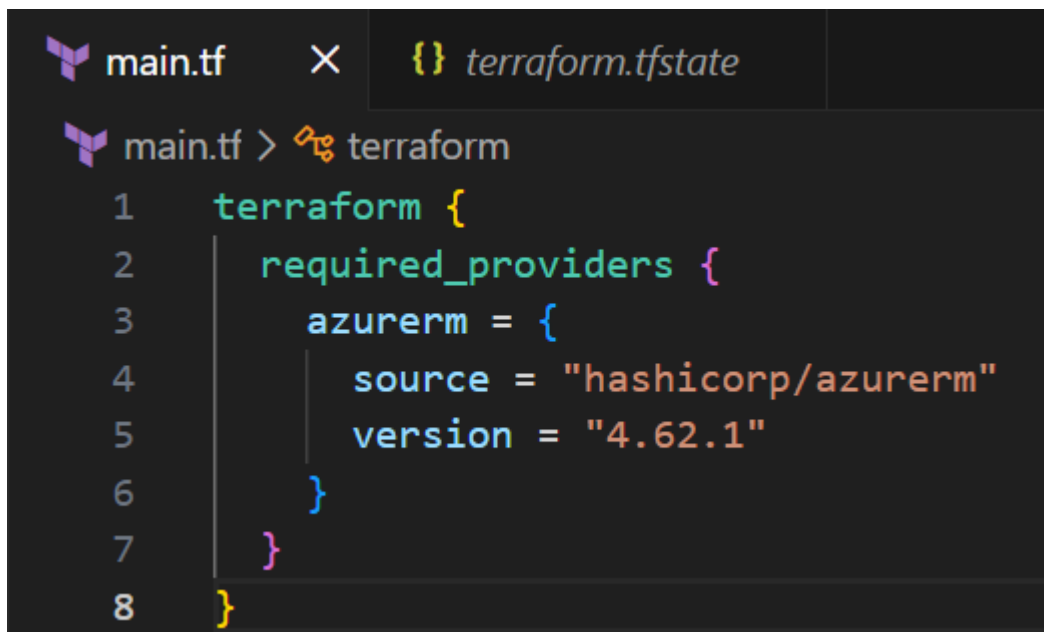
- ✓ Create folder in **C-drive** and open terminal there, Enter **code** .
- ✓ It will **navigate to VS Code**.
- ✓ In **VS code**, click on **folder** and create file (**main.tf**)
- ✓ Inside **main.tf** start creating resources by code.

```
PS C:\AzureTerraform> code .
PS C:\AzureTerraform> terraform --version
Terraform v1.14.6
on windows_amd64
```

AzureRM Provider

The last major release for the AzureRM provider was in ...

- ✓ Within organization we use Stateful Version, if any new updates it will be stateful
- ✓ Every time if new version available we have to test every time.
- ✓ Below 6.34.0 is version if we use it only takes this version so for now, we are not using version.



```
main.tf  X  {} terraform.tfstate

main.tf > terraform
1  terraform {
2      required_providers {
3          azurerm = {
4              source = "hashicorp/azurerm"
5              version = "4.62.1"
6          }
7      }
8  }
```

✓ Below give provider & give Authentication details shown below.

```
main.tf > provider "azurerm" > abc tenant_id
1  #azure provider
2  provider "azurerm" {
3      # Configuration options
4      features {}
5      resource_provider_registrations = "none"
6
7      client_id      = "557622de-75e5-4ba8-9e01-f3d8a2542058"
8      client_secret  = "nQh8Q~kZnCsIhnwFUEFQAR-wcco7NiA.gQ6giaEN"
9      tenant_id     = "4a16e6d8-97ae-4ea2-8fb9-c9aa52ccc600"
10     subscription_id = "c4bc06e7-05d4-47ea-a63f-7a136548588c"
11 }
```

✓ To create Authentication, create App registration on console and give owner rule.

 Uday-app  ...

Search  <<  Deactivate  Delete  Endpoints  Preview features

Overview

- Quickstart
- Integration assistant
- Diagnose and solve problems
- > Manage
- > Support + Troubleshooting

Essentials

Display name : [Uday-app](#)

Application (client) ID : 557622de-75e5-4ba8-9e01-f3d8a2542058

Object ID : aa2993d2-e228-45ff-8cc6-82b2a852d264

Directory (tenant) ID : 4a16e6d8-97ae-4ea2-8fb9-c9aa52ccc600

Supported account types : [My organization only](#)

✓ Client secret

Certificates & secrets

A secret string that the application uses to prove its identity when requesting a token. Also can be referred to as application password.

[+ New client secret](#)

Description	Expires	Value	Secret ID
uday-secrets	02/06/2026	nQh*****	71230827-5eb7-4be2-9837-d170b642b0...  

✓ Subscription id

Subscription name 	Subscription ID 	My role 
Azure subscription 1	c4bc06e7-05d4-47ea-a63f-7a136548588c	Owner

✓ **Next give owner (privilege) to User (Uday-app)**

Display name ↑↓	Application (client) ID	Created on ↑↓	Certificates & secrets
 Uday-app	557622de-75e5-4ba8-9e01-f3d8a2542058	04/03/2026	 Current

✓ **Now the Resources we create using VS code, these resources will create on azure portal by using Authentication & Authorisation.**

✚ **Below I want to create Resource Group, lets see Terraform command and installation on console.**

```
main.tf > provider "azurerm" > abc tenant_id
1  #azure provider
2  provider "azurerm" {
3      # Configuration options
4      features {}
5      resource_provider_registrations = "none"
6
7      client_id      = "557622de-75e5-4ba8-9e01-f3d8a2542058"
8      client_secret  = "nQh8Q~kZnCsIhnwFUEFQAR-wcco7NiA.gQ6giaEN"
9      tenant_id     = "4a16e6d8-97ae-4ea2-8fb9-c9aa52ccc600"
10     subscription_id = "c4bc06e7-05d4-47ea-a63f-7a136548588c"
11 }
12
13 #resourcegroup
14 resource "azurerm_resource_group" "Urg01" {
15     name      = "Uday-RG01"
16     location = "Central US"
17 }
```

- ✓ Open terminal and enter command **terraform init**
- ✓ **Terraform init**, it will refresh the configuration
- ✓ Make sure put Auto Save in VS Code.

```
PS C:\AzureTerraform> code .
PS C:\AzureTerraform> terraform --version
Terraform v1.14.6
on windows_amd64
PS C:\AzureTerraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/azurerm...
- Installing hashicorp/azurerm v4.62.1...
- Installed hashicorp/azurerm v4.62.1 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

✓ Next terraform validate

```
PS C:\AzureTerraform> terraform validate
Success! The configuration is valid.
```

✓ Terraform plan

```
PS C:\AzureTerraform> terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# azurerm_resource_group.Urg01 will be created
+ resource "azurerm_resource_group" "Urg01" {
  + id          = (known after apply)
  + location    = "centralus"
  + name        = "Uday-RG01"
}

Plan: 1 to add, 0 to change, 0 to destroy.
```

✓ Terraform apply –auto-approve

```
PS C:\AzureTerraform> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# azurerm_resource_group.Urg01 will be created
+ resource "azurerm_resource_group" "Urg01" {
  + id          = (known after apply)
  + location    = "centralus"
  + name        = "Uday-RG01"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

azurerm_resource_group.Urg01: Creating...
azurerm_resource_group.Urg01: Still creating... [00m10s elapsed]
azurerm_resource_group.Urg01: Still creating... [00m20s elapsed]
azurerm_resource_group.Urg01: Still creating... [00m30s elapsed]
azurerm_resource_group.Urg01: Creation complete after 30s [id=/subscriptions/c4bc06e7-05d4-47ea-a63f-7a136548588c/resourceGroups/Uday-RG01]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

✓ Below RG created on console

Resource Manager | Resource groups ... [How to manage changes with deployment tools?](#) [Export resource groups using Bicep or Terraform](#) [Summarize my cos](#)

Default Directory

Search ◇ « + Create ⚙️ Manage view ↻ Refresh 📄 Export to CSV 🔗 Open query | 🏷️ Assign tags + Add to service group ⌵

🔍 Filter for any field... Subscription equals all Location equals all ✕ + Add filter

☐	Name ↑	Subscription	Location
☐	AzureBackupRG_southafricanorth_1	... Azure subscription 1	South Africa North
☐	bob-rg	... Azure subscription 1	Central India
☐	NetworkWatcherRG	... Azure subscription 1	Central US
☐	sivarg01	... Azure subscription 1	Central India
☒	Uday-RG01	... Azure subscription 1	Central US

✚ Below I have created more resources

- VNET

```
#virtualNetwork
resource "azurerm_virtual_network" "UVnet" {
  name            = "Uday-Vnet"
  location        = "Central US"
  resource_group_name = "Uday-RG01"
  address_space   = ["10.0.0.0/16"]
}
```

- SUBNET

```
#subnet
resource "azurerm_subnet" "Usubnet" {
  name            = "Uday-subnet01"
  resource_group_name = "Uday-RG01"
  virtual_network_name = "Uday-Vnet"
  address_prefixes   = ["10.0.1.0/24"]
}
```

- NIC

```
#NetworkInterfaceCard
resource "azurerm_network_interface" "UNIC" {
  name            = "Uday-nic"
  location        = "Central US"
  resource_group_name = "Uday-RG01"

  ip_configuration {
    name            = "internal"
    subnet_id       = azurerm_subnet.Usubnet.id
    private_ip_address_allocation = "Dynamic"
    public_ip_address_id = azurerm_public_ip.UdayacceptanceTestPublicIp1.id
  }
}
```


- VM

```
#VirtualMachine
resource "azurerm_virtual_machine" "UVm" {
  name                = "Uday-VM01"
  location            = "Central US"
  resource_group_name = "Uday-RG01"
  network_interface_ids = [azurerm_network_interface.UNIC.id]
  vm_size             = "Standard_D2s_v3"

  # Uncomment this line to delete the OS disk automatically when deleting the VM
  # delete_os_disk_on_termination = true

  # Uncomment this line to delete the data disks automatically when deleting the VM
  # delete_data_disks_on_termination = true

  storage_image_reference {
    publisher = "Canonical"
    offer     = "0001-com-ubuntu-server-jammy"
    sku       = "22_04-lts"
    version   = "latest"
  }
  storage_os_disk {
    name          = "myosdisk1"
    caching       = "ReadWrite"
    create_option = "FromImage"
    managed_disk_type = "Standard_LRS"
  }
  os_profile {
    computer_name = "computername"
    admin_username = "azureuser"
    admin_password = "Password@123"
  }
  os_profile_linux_config {
    disable_password_authentication = false
  }
}
```

- NSG

```
#NetworkSecurityGroup
resource "azurerm_network_security_group" "UNsg" {
  name                = "Uday-nsg"
  location            = "Central US"
  resource_group_name = "Uday-RG01"
  security_rule {
    name                = "UNsg"
    priority            = 100
    direction          = "Inbound"
    access             = "Allow"
    protocol            = "Tcp"
    source_port_range  = "*"
    destination_port_range = "*"
    source_address_prefix = "*"
    destination_address_prefix = "*"
  }
}
```

- NSG - Associate - NIC

```
#NSG Associate with NIC
resource "azurerm_network_interface_security_group_association" "NSGwithNIC" {
  network_interface_id      = azurerm_network_interface.UNIC.id
  network_security_group_id = azurerm_network_security_group.UNsg.id
}
```

- Create Public IP

```
#Public IP
resource "azurerm_public_ip" "UdayacceptanceTestPublicIp1" {
  name                = "UdayacceptanceTestPublicIp1"
  resource_group_name = "Uday-RG01"
  location            = "Central US"
  allocation_method   = "Static"
}
```

- After creating Public IP associate PubIP to NIC level for that use below line and add NIC level. **Check Above** at NIC.

```
public_ip_address_id = azurerm_public_ip.UdayacceptanceTestPublicIp1.id
```